

Physics-Informed Neural Network for Nonlinear Dynamics in Fiber Optics

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A physics-informed neural network (PINN) that combines deep learning with physics is studied to solve the nonlinear Schrödinger equation for learning nonlinear dynamics in fiber optics. A systematic investigation and comprehensive verification on PINN for multiple physical effects in optical fibers is carried out, including dispersion, self-phase modulation, and higher-order nonlinear effects. Moreover, both the special case (soliton propagation) and general case (multipulse propagation) are investigated and realized with PINN. In existing studies, PINN is mainly effective for a single scenario. To overcome this problem, the physical parameters (i.e., pulse peak power and amplitudes of subpulses) are hereby embedded as additional input parameter controllers, which allow PINN to learn the physical constraints of different scenarios and perform good generalizability. Furthermore, PINN exhibits better performance than the data-driven neural network using much less data, and its computational complexity (in terms of number of multiplications) is much lower than that of the split-step Fourier method. The results show that PINN is not only an effective partial differential equation solver, but also a prospective technique to advance the scientific computing and automatic modeling in fiber optics.

complementary ideas, and methods from many different fields of mathematics are essential.^[2] Moreover, modeling and forecasting the dynamics of multiphysics and multiscale nonlinear systems remains an open scientific problem, including but not limited to physical, chemical, or biological systems. Therefore, an extensive investigation with deeper analysis, and more efficient modeling on the complex nonlinear dynamic systems is particularly important. To well explain and characterize the nonlinear dynamics, decades of extensive explorations have made tremendous progress in diverse fields, such as the nonlinear vibration problem in nonlinear mechanics,^[3] three-body problem in celestial mechanics,^[4] and atmospheric turbulence problem in meteorology^[5] by solving complex partial differential equations (PDEs) in these problems. Generally, these PDEs do not have analytical solutions, and numerical methods such as the finite-difference method, finite element method, as well as spectral and

meshless methods have made it possible to obtain approximate solutions. Despite great progress, these conventional numerical methods still have difficulties in solving physical problems with missing initial and boundary conditions (e.g., inferring the initial pulse from the received pulse). For solving inverse problems (e.g., inferring the physical parameters of PDE from limited data), numerical methods are often prohibitively expensive or even impossible. In addition, strict mathematical theories and expert knowledge are required for application of these techniques.

In fiber optics, nonlinear dynamics is the physical principle of fiber-based optical devices, optical information processing, photonic material design, and optical signal transmission, which has greatly promoted the development of fiber lasers, fiber amplifiers, fiber waveguides, and fiber-optic communications.^[6–10] For a fully understanding of the nonlinear dynamics in optical fibers, the propagation dynamics of optical pulses through a fiber can be governed by a fundamental nonlinear PDE, i.e., nonlinear Schrödinger equation (NLSE),^[11] which cannot be solved analytically when dispersion and nonlinearity coexist. To solve the NLSE, the split-step Fourier method (SSFM) and its modifications have been applied for studying various nonlinear effects in optical fibers, mainly due to its straightforward

1. Introduction

Generally, the dynamics of nonlinear systems must be described by nonlinear models; yet, their theory is technically difficult and detailed analytical solutions are not always obtainable.^[1] To understand some aspects of such nonlinear dynamics, various

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implementation and relative accuracy compared with other numerical methods.^[12,13] However, for long-distance scenarios with high nonlinearity, the step lengths of SSFM must be reduced considerably to meet the accuracy requirement, which increases the computational complexity. Moreover, when the NLSE is solved for high-dimensional problems (e.g., simulation of wavelength-division-multiplexed systems), the temporal resolution should be a small fraction of the entire bandwidth. This results in extremely large mesh points sets in the time domain, and thus, becomes a quite time-consuming process. Hence, it is desirable to develop more powerful and advanced techniques to directly solve the NLSE without complex numerical calculations.

In the past decade, deep learning (DL) has evolved rapidly and studied extensively to solve all sorts of issues, including various nonlinear dynamic systems.^[14,15] At the beginning, DL was mainly applied as a black-box tool based on the data-driven concept without considering any prior knowledge. Given that the input–output data pairs are available for training, this black-box method for data fitting learns the input–output relationship of an analytically intractable system in a data-driven manner. To date, various nonlinear problems in optical fibers have been solved using DL with different neural network structures based on very large datasets collected from either experiments or simulations, bridging the nonlinear mapping between input parameters and output targets.^[16–19] However, although this black-box approach can reduce the computational complexity to a certain extent and expand the types of problems that can be solved, its performance is heavily reliant on the quantity and quality of data. Moreover, data-driven networks are usually trained without considering the underlying mathematics and physics. However, such prior knowledge is essential for fully understanding the philosophy behind nonlinear dynamics.

Recently, a physics-informed neural network (PINN) has been proposed,^[20,21] which fully considers prior knowledge, including essential mathematical equations, physical theories, and corresponding constraint conditions of target problems. Despite newly proposed, PINN has attracted widespread attention and been verified in multiple nonlinear dynamic systems governed by different PDEs, such as the Navier–Stokes equation of fluid dynamics,^[20,22] Burgers' equation of aerodynamics,^[20,23] and heat transfer equation of thermodynamics.^[24,25] Unlike data-driven neural networks that simply fit their outputs to given inputs, PINN manifests itself as an automatic PDE solver that maps the independent variables of PDE to its solutions. More specifically, the governing equations as well as physical constraints, including the initial and boundary conditions, constitute the network loss function and serve as regularization mechanisms. Hence, PINN transforms the PDE solving problem into a loss function optimization problem. In this manner, the network integrates the prior knowledge of mathematics and physics throughout the entire training process. To calculate the governing equations, the automatic differentiation technique is employed,^[26] and the differential terms of PDE are calculated automatically according to auxiliary coordinates, without the exact data at these coordinates. Hence, the data dependency of the model is greatly reduced. Because the complexity is mainly determined by the network structure, the computational complexity of PINN with stable structure

is almost unchanged for different scenarios, which is quite different from numerical methods. Moreover, PINN has also achieved remarkable results in solving problems with missing physical constraints and inverse problems.^[27,28] Therefore, PINN combines the advantages of both numerical and data-driven methods while overcoming their limitations. Currently, studies on nonlinear dynamics in fiber optics were mainly based on numerical and data-driven methods, and the existing PINN-based work was preliminary in solving NLSE.^[29,30] A systematic investigation and comprehensive verification on this subject is yet to be undertaken. Additionally, the current PINN models achieve poor generalizability and cannot be directly applied when the physical constraints change without remodeling. Therefore, it is still necessary to separately establish models for different physical constraints in the study of nonlinear dynamics involving multiple scenarios in fiber optics, which greatly limits the application value of PINN.

In this study, we propose a PINN-based solution to characterize complex nonlinear dynamics and model multiple physical effects in fiber optics. The general context of our work is the recent development of the combination of DL and physics, which solves complex PDEs with prior knowledge involving the governing equations and physical constraints of nonlinear dynamic systems. To improve the generalizability of PINN, the physical parameters (i.e., pulse peak power and amplitudes of subpulses) are embedded to the PINN input layer as physical parameter controllers, thereby allowing the PINN model to learn different physical constraints and simulate the corresponding physical scenarios. We carried out a systematic investigation and comprehensive verification on multiple physical effects in optical fibers, including chromatic dispersion, self-phase modulation (SPM), and higher-order nonlinear effects. Moreover, both soliton propagation and multipulse propagation were investigated and realized with PINN. The accuracy and complexity of the PINN model are also studied in this work and results show that PINN exhibits better performance than data-driven methods with much less data, and the computational complexity (in terms of multiplications) is usually two orders of magnitude lower than that of SSFM. Therefore, PINN is not only equivalent to a complex operator for solution of PDEs with various physical constraints, but can also act as a supplementary tool to advance the theoretical understanding of nonlinear dynamics in fiber optics.

2. Concept and Principles

Multiple linear and nonlinear effects occur in optical fibers, causing various distortions during pulse propagation, which can be well characterized by NLSE. The NLSE is a complex PDE containing multiple higher-order differential terms. To calculate these differential terms for a given accuracy, more iterations and smaller step size are usually required using numerical methods, which increases the complexity tremendously. Instead of directly calculating the differential terms, the conventional DL methods usually rely on large amounts of simulation or experimental data for modeling. In contrast, PINN not only takes both accuracy and complexity into account, but also makes full use of prior knowledge of underlying mathematics and physics to provide a reliable and efficient solution for complex nonlinear dynamic systems.

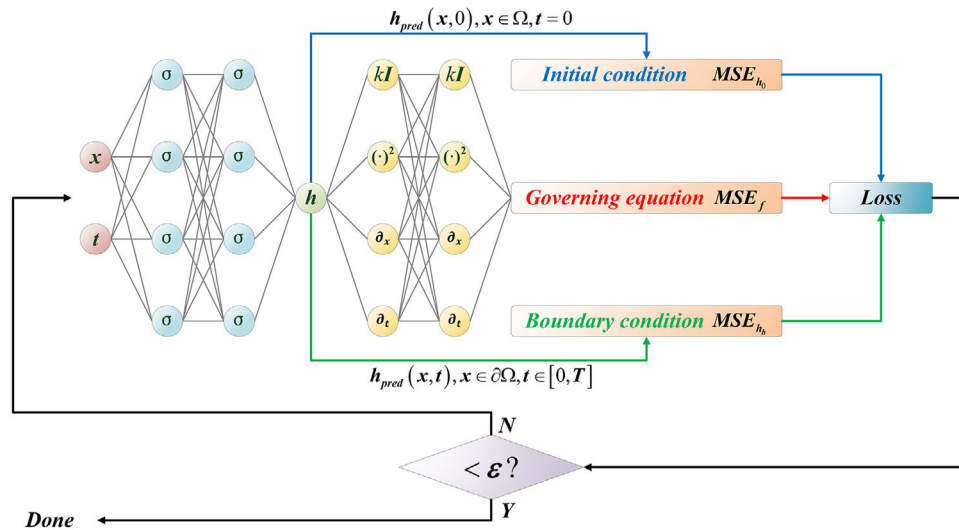


Figure 1. Principle of physics-informed neural network.

Next, we will illustrate the theory of PINN in detail and explain its modeling principles of nonlinear dynamics in fiber optics.

2.1. Overview of PINN

Most PDEs can be expressed in the following general form including governing equation, initial and boundary conditions

$$\begin{aligned} h_t + \mathcal{N}_x(h) &= 0, x \in \Omega, t \in [0, T] \\ h(x, 0) &= h_0(x, 0), x \in \Omega, t = 0 \\ h(x, t) &= h_b(x, t), x \in \partial\Omega, t \in [0, T] \end{aligned} \quad (1)$$

where $h(x, t)$ denotes the solution of PDE, and h_t is the temporal derivative; $\mathcal{N}_x$ is a generalized nonlinear differential operator, including spatial derivatives and other nonlinear terms, such as $|h|^2$. Ω and $\partial\Omega$ denote the computational domain and boundary. Further, $h_0(x, t)$ and $h_b(x, t)$ denote the solutions at $t = 0$ and the boundary, respectively.

To solve the PDE with the initial and boundary conditions given in Equation (1), a PINN composed of a multilayer deep neural network is established to approximate h , as shown in **Figure 1**. The PINN takes the temporal-spatial auxiliary coordinates (x, t) as input and the predicted solution $h(x, t)$ as output. To meet the constraints of the governing equation, it is necessary to accurately calculate the temporal derivatives, spatial derivatives, and the nonlinear terms, so as to estimate the value of the governing equation. With the help of automatic differentiation, the differential and nonlinear terms of h can be calculated through equation parameters and input auxiliary coordinates. Moreover, as a mesh-free method, automatic differentiation does not suffer from errors such as truncation and round-off errors, which are quite common in conventional numerical methods. Furthermore, the solutions at $t = 0$ and the boundary will be predicted and compared with the exact data at these places to satisfy the initial and boundary conditions. Therefore, the network parameters can be learned by minimizing the mean square error (MSE) loss of the

governing equation, initial and boundary conditions, which is expressed as

$$\begin{aligned} \text{Loss} &= \text{MSE}_f + \text{MSE}_{h_0} + \text{MSE}_{h_b} \\ &= \frac{1}{N_f} \sum_{i=1}^{N_f} |h_t + \mathcal{N}_x(h)|^2 + \frac{1}{N_0} \sum_{i=1}^{N_0} |h_{\text{pred}}(x^i, 0) - h_0(x^i, 0)|^2 \\ &\quad + \frac{1}{N_b} \sum_{i=1}^{N_b} |h_{\text{pred}}(x^i, t^i) - h_b(x^i, t^i)|^2 \end{aligned} \quad (2)$$

where MSE_f , MSE_{h_0} , and MSE_{h_b} penalize the residuals of the governing equation, initial and boundary conditions, respectively. N_f , N_0 , and N_b are the numbers of auxiliary coordinates required to calculate each MSE term. $h_{\text{pred}}(x^i, t^i)$ denotes the solution at (x^i, t^i) predicted by PINN. It should be noted that the boundary condition given in Equation (2) is the Dirichlet boundary condition, and the boundary derivative should also be included when the PDE has a mixed boundary condition. When the loss is minimized to a small value ϵ after multiple iterations, the PINN network can be equivalent to a complex operator that solves this PDE.

This advanced technology has achieved great success in fluid dynamics, aerodynamics, thermodynamics, and biomedicine. Thus, it greatly motivates us to explore if PINN can fundamentally reassess nonlinear dynamics in fiber optics from a fresh perspective.

2.2. PINN-Based Modeling for Nonlinear Dynamics in Fiber Optics

For nonlinear dynamics in fiber optics, the NLSE can characterize various physical effects in optical fibers, such as group velocity dispersion (GVD), third-order dispersion (TOD), and SPM. By introducing dimensionless parameters, the NLSE can be simplified

to the following normalized form^[11]

$$h_z + \frac{\alpha}{2}h + \frac{i}{2}\beta_2 h_{tt} - \frac{1}{6}\beta_3 h_{ttt} - iN^2 [|h|^2 h + is(|h|^2 h)_t - \tau_R(|h|^2)_t h] = 0 \quad (3)$$

where the dimensionless parameters are defined as

$$h = \frac{H}{\sqrt{P_0}}, z = \frac{Z}{L_D}, t = \frac{T}{T_0}, N^2 = \frac{L_D}{L_{NL}} = \frac{\gamma P_0 T_0^2}{|\beta_2|}, L_D = \frac{T_0^2}{|\beta_2|}, L_{NL} = \frac{1}{\gamma P_0}, s = \frac{1}{\omega_0 T_0}, \tau_R = \frac{T_R}{T_0} \quad (4)$$

Here, $H(Z, T)$ is the complex envelope of a slowly varying optical field. Z and T are the propagation distance and time scale in a frame of reference moving with the pulse at group velocity, respectively. The dimensionless parameters h , z , and t are the normalized values of H , Z , and T under peak power P_0 , dispersion length L_D , and pulse width T_0 . N^2 is the ratio of L_D to the nonlinear length L_{NL} , which governs the relative impact of GVD and SPM effects on pulse evolution along the fiber. The propagation parameters α , β_2 , β_3 , and γ reflect the power attenuation, GVD, TOD, and strength of nonlinear effects during the pulse evolution, respectively. For ultrashort optical pulses with $T_0 < 1$ ps, higher-order nonlinear effects such as self-steepening (SS)^[31] and intrapulse Raman scattering (IRS)^[32] as expressed by the last two terms of Equation (3), should be considered, where s and τ_R are the normalized parameters of SS and IRS, respectively, while ω_0 and T_R in Equation (4) are the central angular frequency of the pulse and first moment of Raman response function, respectively.

For optical fibers, the physical constraint is mainly the initial pulse, which directly influences the evolution characteristics. As the NLSE is a complex-valued equation, it is advisable to divide this governing equation into real- and imaginary-parts, so as to allow the PINN to be applicable to optical fibers. Thus, we write $h(z, t)$ in rectangular form as $h(z, t) = u(z, t) + iv(z, t)$, where $u(z, t)$ and $v(z, t)$ are the real and imaginary parts of $h(z, t)$. Accordingly, the NLSE in Equation (3) can be rewritten with separate real- and imaginary- parts in the following form

$$f(u, v) + ig(u, v) = 0 \quad (5)$$

where f and g are the real- and imaginary- parts functions of the NLSE, and are expressed as

$$\begin{aligned} f : u_z + \frac{\alpha}{2}u - \frac{1}{2}\beta_2 v_{tt} - \frac{1}{6}\beta_3 u_{ttt} + N^2 \{ (u^2 + v^2) v \\ + s[(u^2 + v^2) u]_t - \tau_R(u^2 + v^2)_t v \} \\ g : v_z + \frac{\alpha}{2}v + \frac{1}{2}\beta_2 u_{tt} - \frac{1}{6}\beta_3 v_{ttt} - N^2 \{ (u^2 + v^2) u \\ - s[(u^2 + v^2) v]_t - \tau_R(u^2 + v^2)_t u \} \end{aligned} \quad (6)$$

Considering both the governing equations and initial condition, the loss function of PINN for nonlinear dynamics in fiber

optics can be expressed as

$$\begin{aligned} Loss = MSE_{u_0} + MSE_{v_0} + MSE_f + MSE_g \\ = \frac{1}{N_0} \sum_{i=1}^{N_0} |u_{\text{pred}}(0, t^i) - u_0(0, t^i)|^2 \\ + \frac{1}{N_0} \sum_{i=1}^{N_0} |v_{\text{pred}}(0, t^i) - v_0(0, t^i)|^2 \\ + \frac{1}{N_f} \sum_{i=1}^{N_f} |f(z^i, t^i)|^2 + \frac{1}{N_f} \sum_{i=1}^{N_f} |g(z^i, t^i)|^2 \end{aligned} \quad (7)$$

where MSE_{u_0} and MSE_{v_0} correspond to the real- and imaginary-part constraints of the initial pulse, while MSE_f and MSE_g penalize training results that do not satisfy the constraints of governing equations. Further, $u_0(0, t^i)$ and $v_0(0, t^i)$ denote the real and imaginary parts of the initial pulse, respectively; $u_{\text{pred}}(0, t^i)$ and $v_{\text{pred}}(0, t^i)$ are the predicted outputs of real and imaginary parts with input $(0, t^i)$, respectively, and $f(z^i, t^i)$ $g(z^i, t^i)$ are the results of Equation (6) calculated by PINN with input (z^i, t^i) . N_0 and N_f are the numbers of auxiliary coordinates required to calculate the MSE terms of initial conditions, and governing equations.

Figure 2 shows the detailed process of modeling nonlinear dynamics in fiber optics using PINN. As shown in Figure 2a, a temporal-spatial 2D region is constructed to study the evolution dynamics of pulses in fiber, and the exact evolution data of various pulses can be generated by SSFM as a reference. To meet the constraints of initial condition and governing equations, N_0 data points of the initial pulse and N_f auxiliary coordinates in the 2D region are sampled, as shown in Figure 2b. It is notable that the exact value at these sampled auxiliary coordinates is unknown. Following the PINN network and automatic differentiation shown in Figure 2c, the temporal-spatial coordinates (z, t) are used to calculate the four MSE terms in Equation (7), and the nonlinear dynamics in fiber optics can be well characterized after multiple iterations.

3. Demonstrations and Results

In the modeling process described above, a 2D region of $[-24T_0, 24T_0] \times [0, 4L_D]$ was constructed to reveal the pulse evolution. For typical standard single-mode fibers, β_2 is ~ -20 ps km⁻¹, T_0 is ~ 10 – 100 ps, and L_D is ~ 5 – 500 km, according to Equation (4). Following the space-filling Latin hypercube sampling strategy,^[33] $N_f = 50\,000$ auxiliary coordinates were sampled for calculating MSE_f and MSE_g terms. The other two MSE terms which govern the initial condition (i.e., MSE_{u_0} and MSE_{v_0}) were calculated with $N_0 = 256$ coordinates uniformly sampled on the initial pulse and the corresponding real and imaginary parts. Generally, a PINN with more hidden layers and neurons per layer shows better model performance. However, after trial and error, when the number of neurons per layer exceeds 80, it has little effect on model performance, and the larger storage space and longer training time lead to minor improvement in model performance when the number of hidden layers exceeds 4. Furthermore, the optimal network structure for different problems is not always the same.

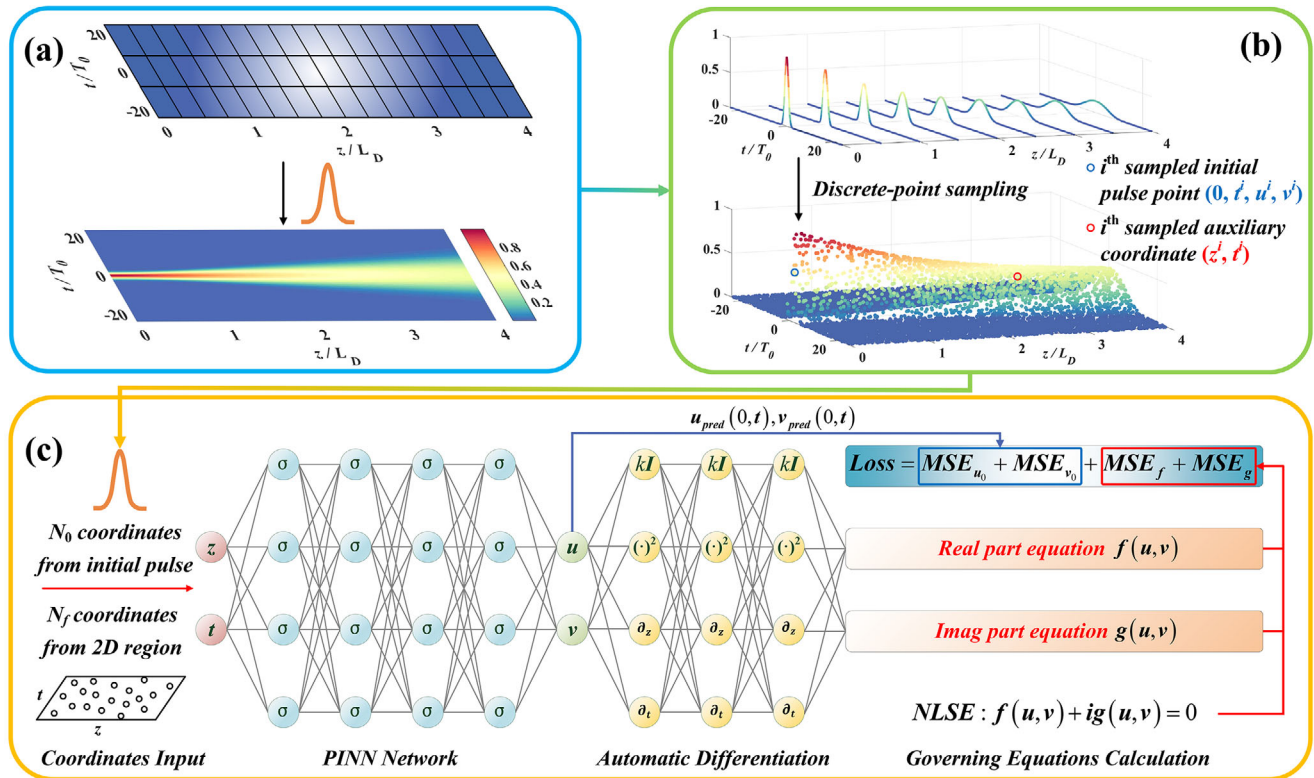


Figure 2. Process of modeling nonlinear dynamics in fiber optics using PINN: a) temporal-spatial 2D region and exact evolution data generated by SSFM as a reference, b) discrete-points sampling for PINN, and c) the PINN structure constructed for solving the NLSE in fiber.

Therefore, to achieve a reasonable tradeoff between accuracy and complexity, we comprehensively considered a PINN network consisting of four hidden layers with 100 neurons and activated by the tanh function to characterize all test scenarios in this work. Moreover, compared with gradient descent-based optimizers such as Adam, the limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) optimization algorithm approximates the second-order Hessian matrices of the loss function, which can find a more accurate direction of gradient descent.^[34,35] Therefore, to make the network converge faster with less fluctuation, the L-BFGS optimization algorithm is adopted to accelerate the PINN learning in this work. Specifically, the L-BFGS optimization ends when the $(k+1)^{th}$ and k^{th} iterations satisfied Equation (8)

$$\frac{Loss^k - Loss^{k+1}}{\max\{|Loss^k|, |Loss^{k+1}|, 1\}} < \epsilon \quad (8)$$

3.1. Multiple Physical Effects in Fiber Modeled by PINN

3.1.1. Dispersion and SPM Coexisting

In a typical optical fiber communication system, L_D and L_{NL} are generally shorter than fiber length L . Thus, the pulse propagation in fiber is mainly affected by dispersion and SPM, which are the two fundamental effects in fiber optics. For this general case, f

and g should involve the terms that govern the dispersion and SPM.

Figure 3a,b shows the evolution of the Gaussian pulse in the normal dispersion regime ($\beta_2 > 0$) when GVD and SPM effects coexist. In this case, SPM will accelerate the dispersion-induced pulse broadening. This behavior can be explained by recognizing that the SPM generates red- and blueshifted components near the leading and trailing edges of the pulse, and the redshifted component moves faster in the normal dispersion regime. By comparing the evolution results generated by PINN and SSFM, the error density plot representing their absolute error can be obtained, as shown in Figure 3c. It should be noted that the error density plots in this paper represent the absolute error of the evolution results generated by PINN and SSFM, which can be expressed as

$$\text{Absolute error} = |h^2(z, t) - h_{pred}^2(z, t)| \quad (9)$$

It can be seen that the absolute error appears larger initially and diverges during pulse propagation. Although the pulse amplitude is significantly low in the presence of SPM, as shown in Figure 3d,e, the PINN-based model can effectively model the pulse evolution of this typical fiber scenario.

Unlike the evolution obtained when GVD and SPM effects exist simultaneously, more details appear at the zero-dispersion wavelength case when TOD and SPM coexist. As shown in Figure 4a,b, oscillation peaks appear on the trailing edge of the pulse during evolution. In addition, nonzero intensity appears

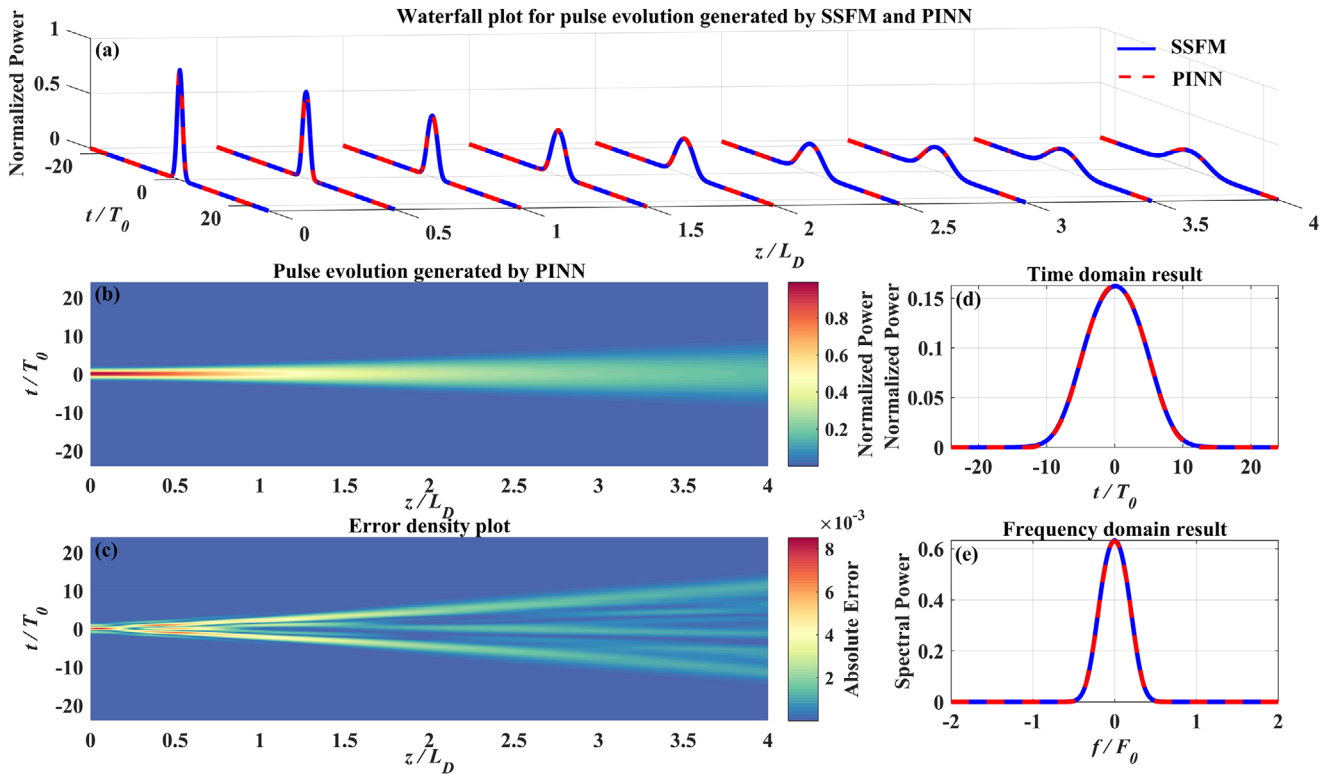


Figure 3. Effects of GVD and SPM on pulse propagation: a) waterfall plot for pulse evolution generated by SSFM and PINN, b) pulse evolution generated by PINN, c) error density plot of PINN compared with SSFM, and d) time- and e) frequency-domain results for normalized and spectral powers at $4L_D$, respectively.

at the oscillation minima, as shown in Figure 4d. From the error density plot of Figure 4c, the absolute error is slightly higher than the previous case; here, the maximum error reached 1.56×10^{-2} . Nevertheless, the evolution under the mixed TOD and SPM action can still be well characterized by PINN. In the optimization process, PINN gradually fitted the main, first oscillation peaks, and so on. From the frequency-domain results shown in Figure 4e, the pulse spectrum has an asymmetric double-peak structure, illustrating that TOD introduces spectral asymmetry without affecting the two-peak structure of the SPM. This effect is in sharp contrast to Figure 3e in which GVD hindered splitting of the spectrum in the normal dispersion regime.

3.1.2. Higher-Order Nonlinear Effects

Thus far, we have verified the PINN-based optical fiber modeling in terms of the mixed impact of linear effects and SPM. Next, the higher-order nonlinear effects such as SS and IRS, which are incorporated for ultrashort optical pulses need to be further investigated. Considering the anomalous dispersion case ($\beta_2 < 0$), and keeping the last two terms in the NLSE, the pulse evolutions under SS and IRS effects are shown in Figures 5 and 6, respectively.

It should be noted that, during modeling of the SS effect, we expanded the distance z to $10L_D$ and set s to 0.2, to improve the visibility of pulse shift during evolution. Under the SS effect,

the peak moves more slowly than the two sides, as shown in Figure 5a,b, and appears shifted toward the trailing side in the anomalous dispersion regime. In particular, the trailing edge becomes steeper with increasing z , if anomalous dispersion is neglected. As the propagation progresses, the pulse shape gradually stabilizes, which is more obvious when the initial pulse is a soliton-like pulse $\text{sech}(t)$. As shown in Figure 5c–e, the error is relatively large and mainly concentrated near the peak in this case; this illustrates that the approximation of higher-order differential terms is not easy. In fact, more auxiliary coordinates and exact data can improve model performance, which will be discussed further in Section 4.1.

For ultrashort optical pulses, IRS becomes quite important. Thus, we set $\tau_R = 0.1$ and $N = 1.4$ to better reflect the effects of IRS on pulse evolution. As shown in Figure 6a,b, even if τ_R is small, it induces pulse fission. As in the previous case, the pulse fission is more obvious when a soliton is transmitted. Moreover, the pulse exhibits a large temporal shift in the time domain and a Raman-induced frequency shift toward longer wavelengths in the frequency domain. These two phenomena are direct consequences of the IRS. From the pulse evolution shown in Figure 6a,b, the pulse decelerates after a narrowing process and exhibits a curved trajectory owing to the slower group velocity at longer wavelengths.

These results indicate that, for the same model configuration, PINN can model these higher-order nonlinear effects effectively. Furthermore, there is only a minimal increase in computational

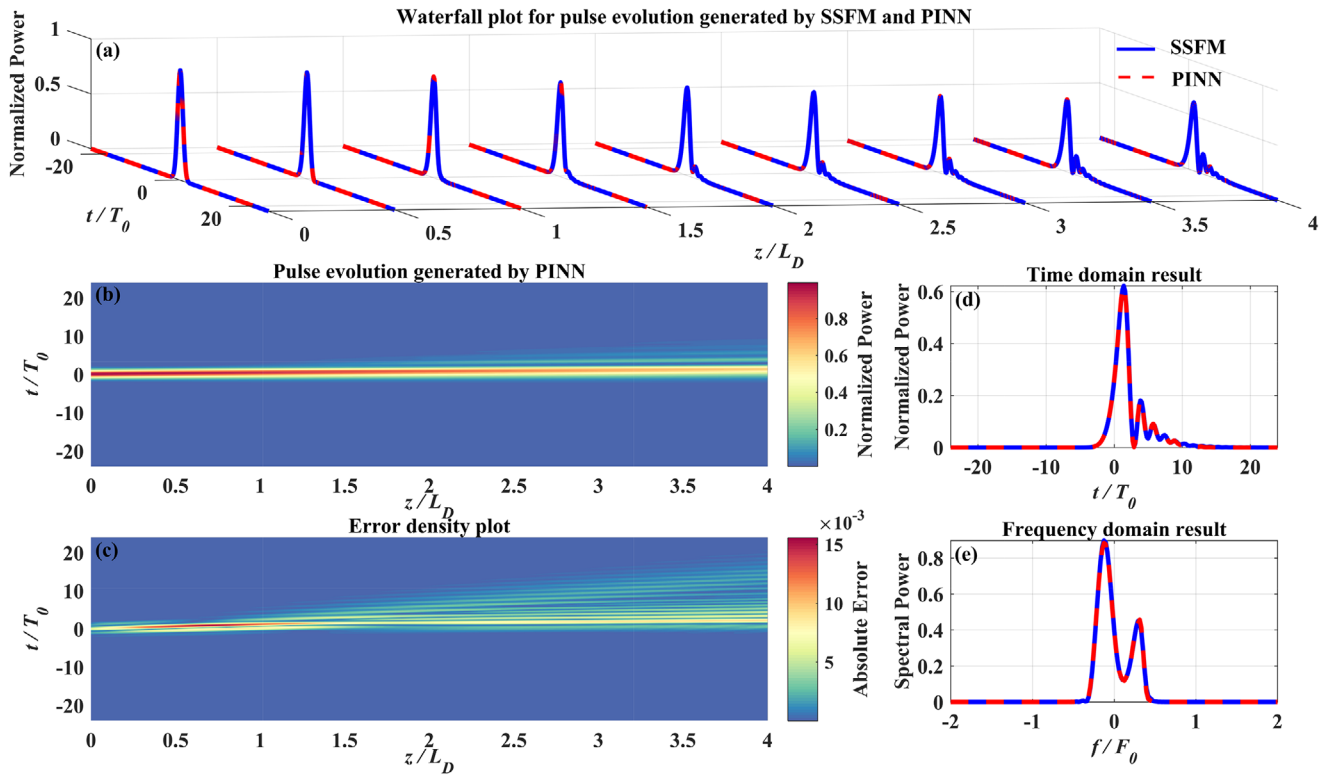


Figure 4. Effects of TOD and SPM on pulse propagation: a) waterfall plot for pulse evolution generated by SSFM and PINN, b) pulse evolution generated by PINN, c) error density plot of PINN compared with SSFM, and d) time- and e) frequency-domain results for normalized and spectral powers at $4L_D$, respectively.

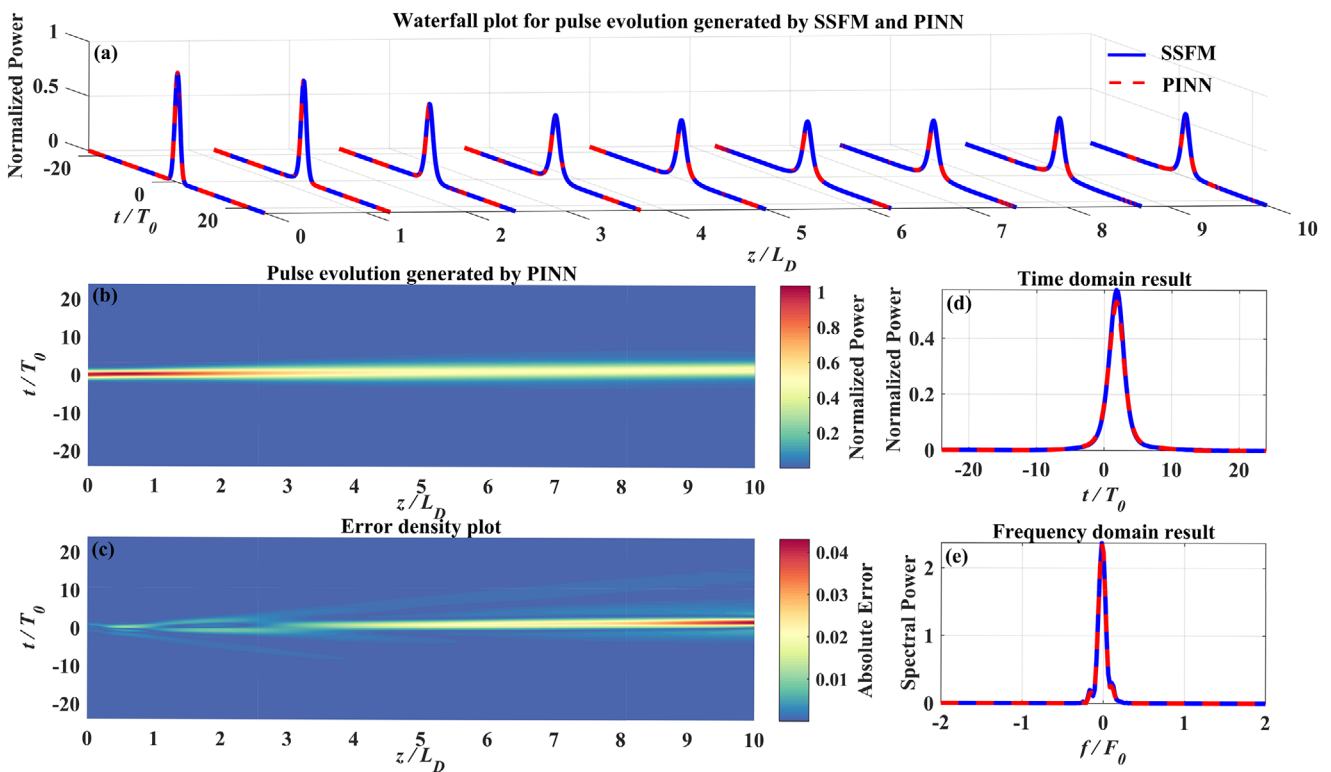


Figure 5. Effects of SS on pulse propagation: a) waterfall plot for pulse evolution generated by SSFM and PINN, b) pulse evolution generated by PINN, c) error density plot of PINN compared with SSFM, and d) time- and e) frequency-domain results for normalized and spectral powers at $10L_D$, respectively.

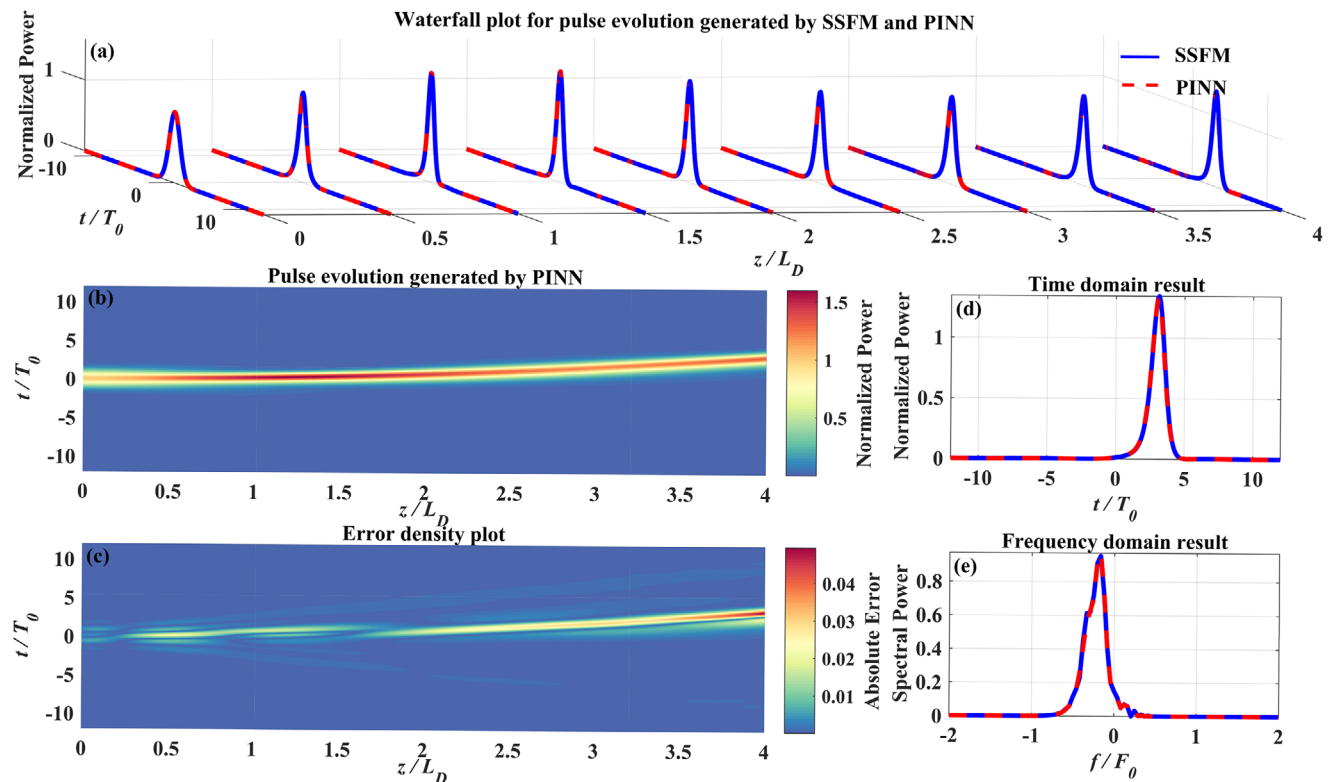


Figure 6. Effects of IRS on pulse propagation: a) waterfall plot for pulse evolution generated by SSFM and PINN, b) pulse evolution generated by PINN, c) error density plot of PINN compared with SSFM, and d) time- and e) frequency-domain results for normalized and spectral powers at $4L_D$, respectively.

complexity because of the unchanged network structure, which is in contrast to numerical methods such as the SSFM. The absolute error of above higher-order nonlinear effects reaches the order of 10^{-2} with PINN, which can be reduced through further optimization. Despite the error, these effects are well characterized by PINN and do not suffer from problems caused by different meshing techniques implemented with numerical methods.

3.2. PINN-Modeling Universality

The above verification results reflect the feasibility of PINN-based modeling of nonlinear dynamics in fiber optics based on single-Gaussian pulses. However, various types of pulses are employed in practical scenarios for different purposes. Next, to verify the universality of PINN modeling method for different waveforms, some more complex and random pulses that are typically employed in optical communications were further studied where both the GVD and SPM effects were considered.

3.2.1. Soliton Propagation

In the anomalous dispersion regime, the interplay between dispersion and nonlinear effects creates modulation instability and induces a pulse-like optical soliton.^[36] This type of soliton usually exhibits a periodic evolution pattern. For the fundamental soliton shown in Figure 7, the waveform remains unchanged during the pulse evolution; this feature renders the soliton

attractive for ultralong distance and ultrahigh capacity optical soliton communication systems. For the setup considered here, the maximum error of the fundamental soliton was 3.9×10^{-3} , as shown in Figure 7c. Clearly, the pulse evolution complexity also affects the model accuracy to a certain extent, apart from the complexity of the governing equations.

The evolution of second-order optical soliton was also studied in this work, as shown in Figure 8. Here, we considered $N = 2$ combined with the initial pulse (i.e., $h(0, t) = 2\text{sech}(t)$), and thus, the governing equations were simplified to the same form as the fundamental soliton. Unlike the fundamental soliton, the second-order soliton presents a periodic evolution pattern with a period of $\pi L_D/2$. As shown in Figure 8c, the error is mainly concentrated in the two peaks because the corresponding pulse power is high, and the influence of pulse power on evolution is further discussed in Section 3.3.1. Except for the slightly lower amplitudes of pulse centers in Figure 8d,e, this periodic case was also well characterized by PINN.

3.2.2. Multipulse Propagation

Apart from the aforementioned single-pulse cases, another essential case in practical application is multipulse propagation in fiber. Here, two complex waveforms composed of five single Gaussian pulses were investigated: a regular waveform with equal amplitudes (Figure 9) and random waveform with nonequal amplitudes (Figure 10). These two waveforms can be

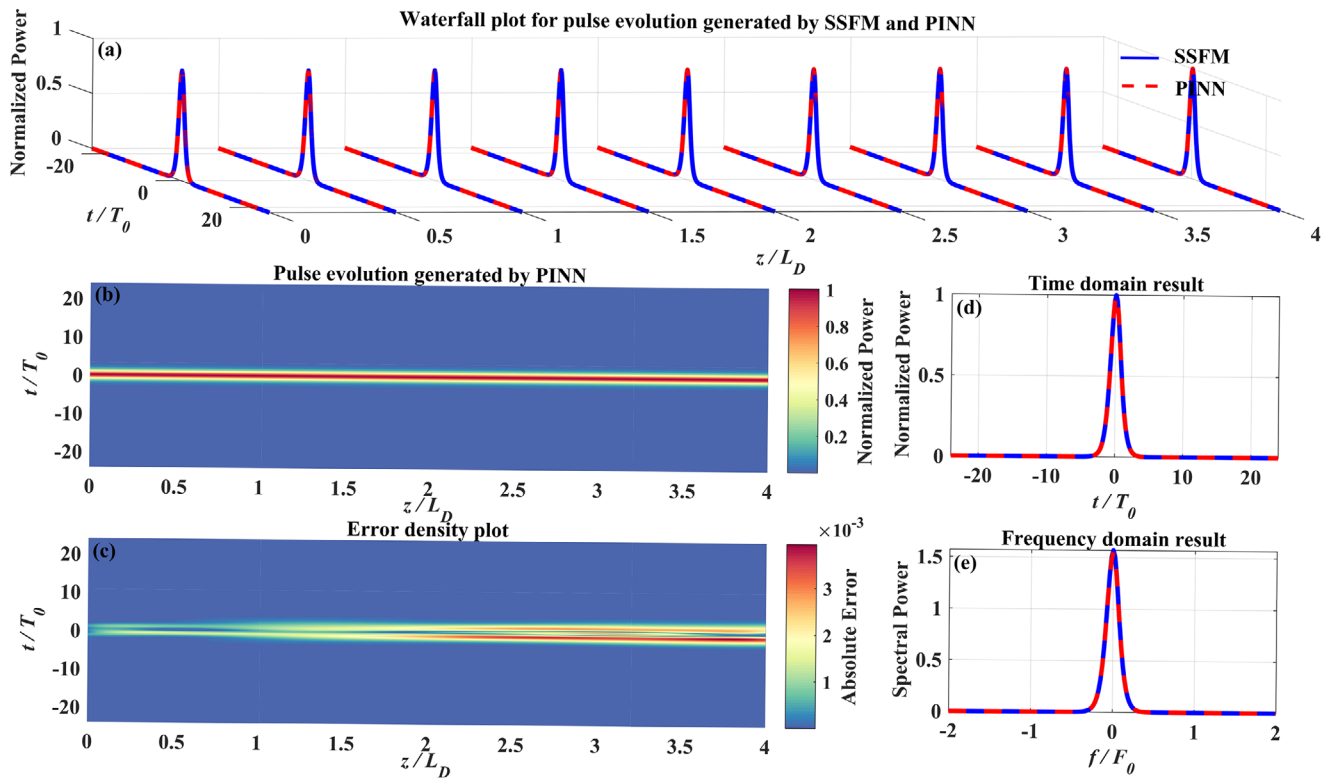


Figure 7. Results for fundamental soliton: a) waterfall plot for pulse evolution generated by SSFM and PINN, b) pulse evolution generated by PINN, c) error density plot of PINN compared with SSFM, and d) time- and e) frequency-domain results for normalized and spectral powers at $4L_D$, respectively.

viewed as two common signals in pulse-amplitude modulation (PAM).

For the equal-amplitude waveform in Figure 9, all single pulses suffer from broadening and crosstalk that occurs among them as the propagation progresses, resulting in newborn peaks between adjacent pulses. For this type of transmission system with interpulse interference, the PINN modeling error reached the level of 10^{-2} for the setup considered here, as shown in Figure 9c. The error is relatively large at the peaks, and especially the central peak, as shown in Figure 9d. This is because the broadening of each pulse is superimposed at these peaks, and the central region is most affected when all pulses have equal amplitudes.

Interpulse crosstalk also appears in the nonequal-amplitude waveform shown in Figure 10. However, the pulse evolution is no longer symmetrical, and the degree of interpulse crosstalk varies with pulse amplitude. Because the initial pulses near the leading edge have lower amplitudes, the error of peaks near the leading edge is relatively large during evolution, as indicated by the error density plot shown in Figure 10d. The maximum error of the time-domain result was 2.16×10^{-2} , which is slightly higher than that of the single-pulse case. For this case where the pulse evolution details are richer, the PINN model can still achieve such good results with the same configuration. Additionally, the model performance can be further improved when more coordinate points and exact data are available. From this preliminary attempt on multipulse investigation, the satisfactory results indicate the considerable potential of this approach for arbitrary waveform modeling.

3.3. Generalizability of PINN Modeling

At present, most PINN-based PDE solutions are still hindered by insufficient generalizability, i.e., a constructed PINN model is only feasible and suitable to one scenario. For different scenarios, the physical constraints change accordingly. Thus, a different optical fiber model must be developed with a new loss function. In previous studies,^[20–25,27–30] the loss function of PINN uniquely corresponds to physical constraints of one scenario in the modeling process, resulting in poor generalizability. However, if multiple physical constraints corresponding to multiple scenarios could be introduced to the loss function during the training process, a multiscenario PINN model could be realized without re-modeling. Here, to verify the feasibility of generalizable PINN model for multiscenario modeling, the optical fiber model was trained under the effects of GVD and SPM using pulses of different powers and random waveforms composed of three sub-pulses, as shown in Figure 11.

3.3.1. Modeling Generalizability for Multipower Scenario in Optical Fiber

As shown in Figure 11, a NLSE-independent variable p representing the initial-pulse peak power is introduced to the input layer to achieve model generalizability. This introduced power variable can be regarded as a physical controller mapping different cases to their own loss function. Specifically, each auxiliary coordinate

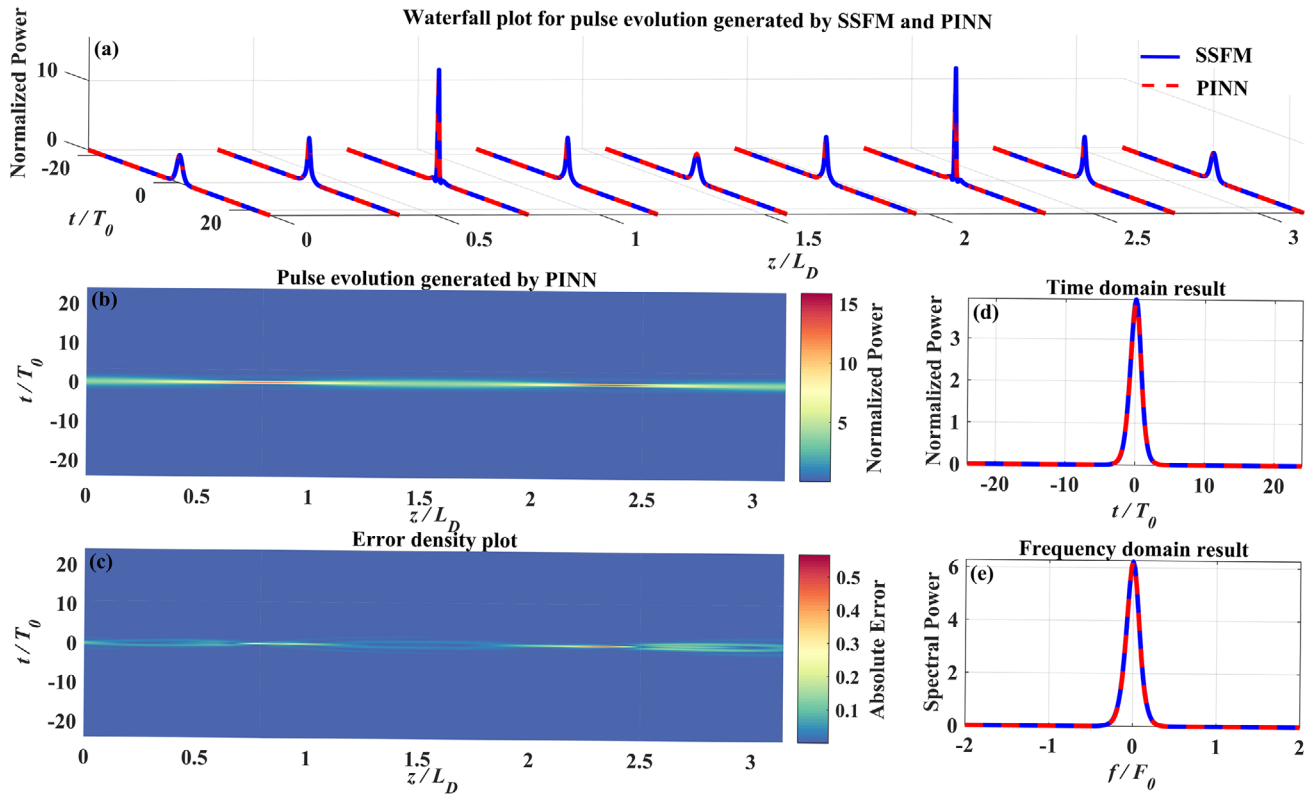


Figure 8. Results for second-order optical soliton: a) waterfall plot for pulse evolution generated by SSFM and PINN, b) pulse evolution generated by PINN, c) error density plot of PINN compared with SSFM, and d) time- and e) frequency-domain results for normalized and spectral powers at πL_D , respectively.

(z^i, t^i) is combined with the peak power p^i . Hence, a new input vector (z^i, t^i, p^i) is formed, with the output consisting of the real and imaginary parts of the complex envelope at (z^i, t^i) with p^i . Each case uniquely corresponds to its specific initial condition, and the new loss function can be expressed in the form of the sum of multiple loss functions, as illustrated in Figure 11.

Next, taking $p = 1$ as a reference, 11 Gaussian pulses with a step size $\Delta p = 0.1$ in the set $[0.5, 1.5]$ were selected as initial pulses. The loss function considered the initial conditions of 6 pulses ($p = [0.5, 0.7, 0.9, 1.1, 1.3, 1.5]$), and the remaining 5 pulses were adopted to verify the power generalizability of PINN. Therefore, the loss function can be written as follows

$$\text{Loss} = \sum_{p \in [0.5, 0.7, 0.9, 1.1, 1.3, 1.5]} [MSE_{u_0}(p) + MSE_{v_0}(p) + MSE_f(p) + MSE_g(p)] \quad (10)$$

In addition, to evaluate the model performance, the relative L_2 error is adopted as a metric, which is calculated with all reference points (201×256) in the 2D region as follows

$$\text{Relative } L_2 \text{ error} = \frac{\sqrt{\sum_{1 \leq i \leq 201, 1 \leq j \leq 256} [h_{\text{pred}}(z^i, t^i) - h(z^i, t^i)]^2}}{\sqrt{\sum_{1 \leq i \leq 201, 1 \leq j \leq 256} h^2(z^i, t^i)}} \quad (11)$$

As shown in **Figure 12a**, the relative L_2 error of each case was less than 1.8×10^{-2} , regardless of whether its corresponding initial pulse was included in the loss function. It can be seen that the error increases with the peak power, which is consistent with the second-order optical soliton case discussed in Section 3.2.1. In fact, this phenomenon is normal because the PINN is essentially a neural network based on a multilayer perceptron and its performance degrades for higher data value, which reflects the importance of data normalization. Moreover, the errors of the 5 verification pulses are all located between adjacent points, indicating the strong generalizability of the PINN model. Figure 12b,c shows the error density plots for $p = 1.4$ and $p = 1.5$, corresponding to cases where the initial pulse constraints are excluded from and included in the loss function, respectively. The error distributions are almost the same, except for a slight difference in amplitude (the error for $p = 1.5$ is greater). This result confirms that the PINN model generalizes the pulse evolution in cases with different powers.

3.3.2. Modeling Generalizability for Random Waveforms in Optical Fiber

The above results illustrate the PINN generalizability in multi-power scenarios. However, random waveform transmission is more general in practice, and even more complicated. Further, we considered transmitted waveforms composed of 3 Gaussian

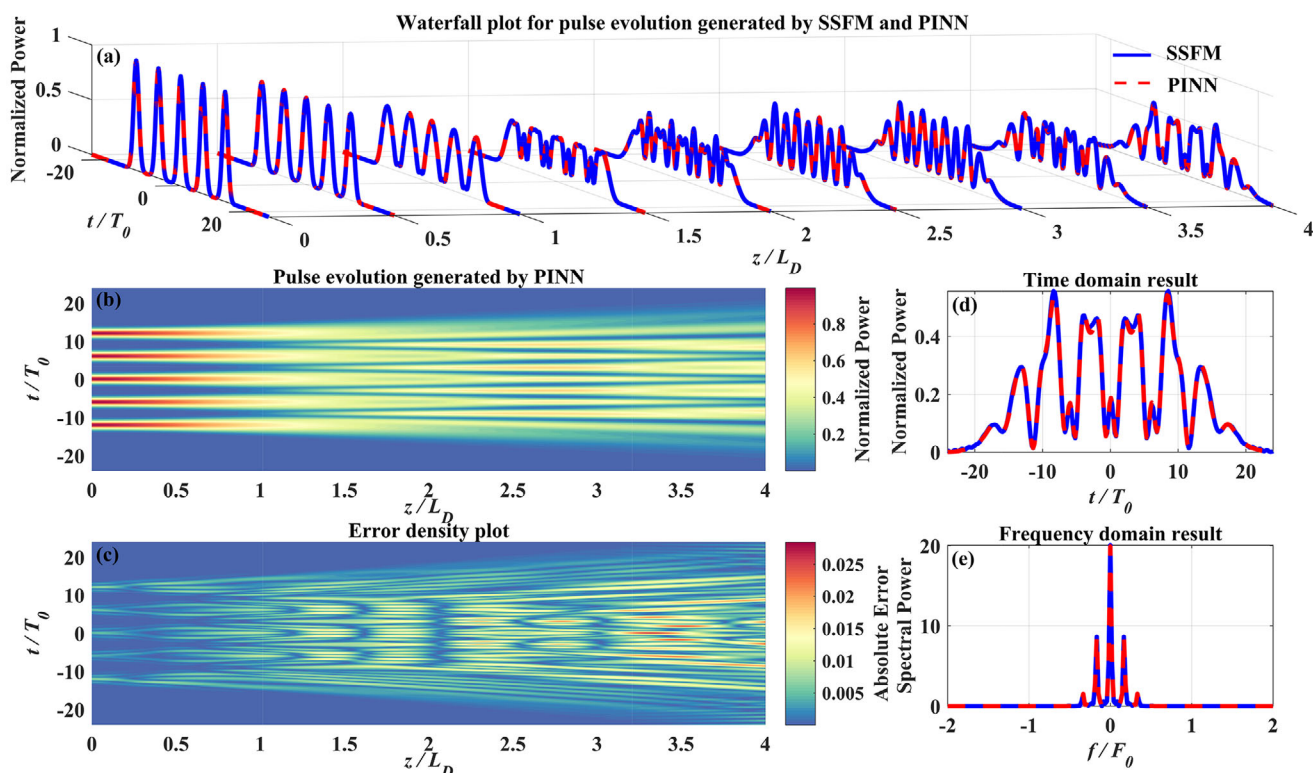


Figure 9. Results for complex waveform of 5 single-Gaussian pulses with equal amplitudes: a) waterfall plot for pulse evolution generated by SSFM and PINN, b) pulse evolution generated by PINN, c) error density plot of PINN compared with SSFM, and d) time- and e) frequency-domain results for normalized and spectral powers at $4L_D$, respectively.

pulses with amplitudes randomly selected from the set [0.25, 0.5, 0.75, 1]. Note that these waveforms could be regarded as PAM4 signals transmitted in an optical fiber communication system. As before, the amplitudes of the 3 subpulses were introduced as physical parameter controllers to the input layer, allowing PINN to learn the evolution of the corresponding waveform via the process shown in Figure 11. Among all possible 3^4 waveforms, 10 random waveforms were selected as the sampled scenarios for PINN to learn. Note that all 4 amplitudes should be included in the selected 10 waveforms to increase model performance. The loss function was still in the sum of multiple loss functions, including the initial constraints of the 10 selected waveforms.

As shown in Figure 13, the PINN-based model effectively characterized the evolution of the two unlearned waveforms. Here, the random waveform composed of 3 subpulses with amplitudes [0.75, 0.5, 0.25] was similar to a waveform composed of 3 subpulses with amplitudes [0.75, 0.75, 0.25] that was learned by the PINN, indicating a relatively small error. In contrast, the random waveform composed of 3 subpulses with amplitudes [0.25, 1, 0.75] differs considerably from the 10 learned waveforms, indicating a relatively large error. Nevertheless, the overall error could be controlled to the order of 10^{-2} . The results for both time and frequency domains are almost consistent with those generated by SSFM (Figure 13c,d,g,h), even though the PINN network never learned those scenarios before. This outcome presents the robust generalizability of PINN, and illustrates that random signals can be effectively learned by PINN. From this perspective, the main significance of PINN is its potential equivalence to a complex op-

erator for solution of complex PDE with arbitrary physical constraints. The PINN-based model established for the above problem can be regarded as a NLSE operator that predicts the evolution of random signals in PAM4 format. Furthermore, PINN has the potential to act as an NLSE operator to characterize random signal evolution in any modulation format.

4. Analysis and Discussions

4.1. PINN-Modeling Accuracy Analysis

A major feature of PINN is to calculate the governing equations through auxiliary coordinates, so that the model meets the constraints of PDE. Therefore, the number of auxiliary coordinates has a great influence on model accuracy. Another factor that affects the accuracy is the strength of physical constraints. Moreover, the model performance will be further improved when physical constraints are stronger. In previous results, only initial conditions were considered. Figure 14 shows the model performance changing with the auxiliary coordinates number under several different physical constraints when GVD and SPM coexist. It can be seen that the model performance increases with the number of auxiliary coordinates and 5000 auxiliary coordinates are enough to acquire a good accuracy for our problem. Moreover, the impact of physical constraints on model performance is intuitive, as shown in Figure 14. The model error will be one order of magnitude higher than the other two cases when only initial pulse is considered. When considering pulses at three distances

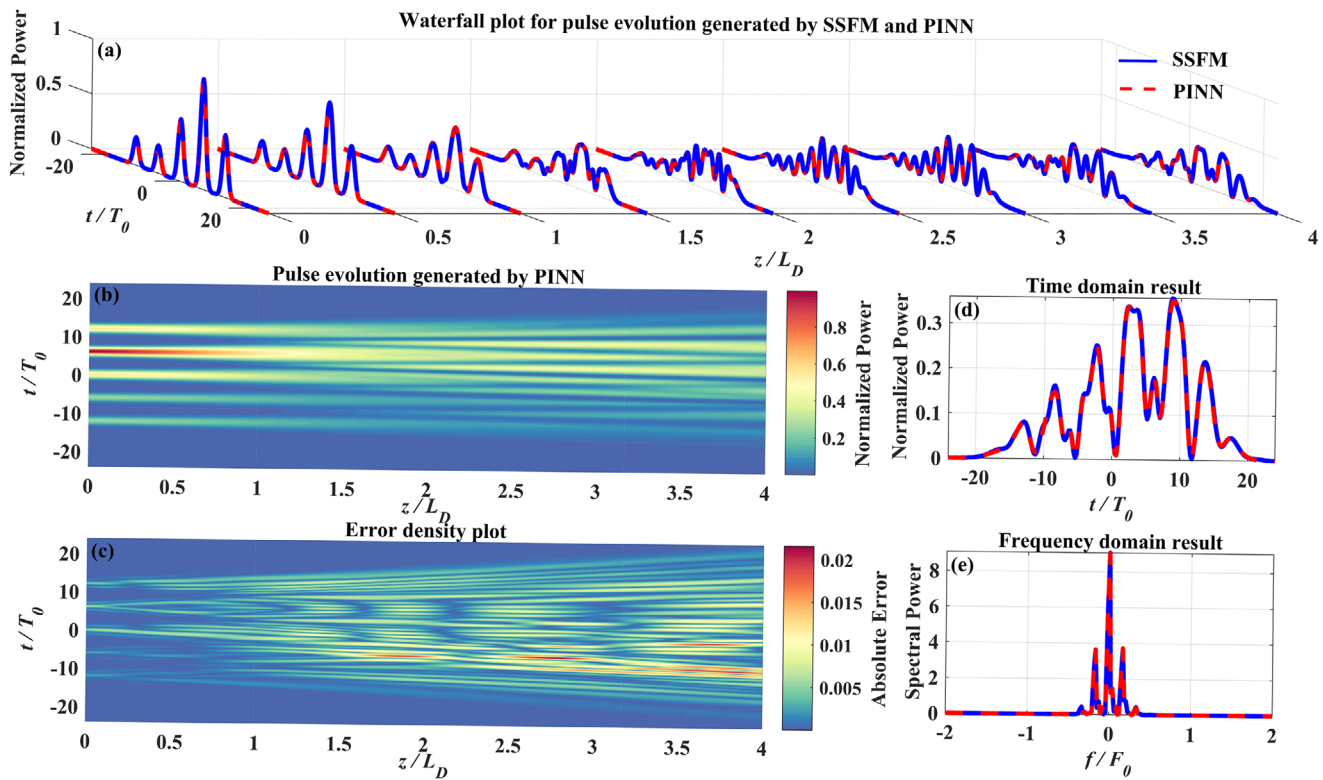


Figure 10. Results for complex waveform of 5 single-Gaussian pulses with nonequal amplitudes: a) waterfall plot for pulse evolution generated by SSFM and PINN, b) pulse evolution generated by PINN, c) error density plot of PINN compared with SSFM, and d) time-, and e) frequency-domain results for normalized and spectral powers at $4L_D$, respectively.

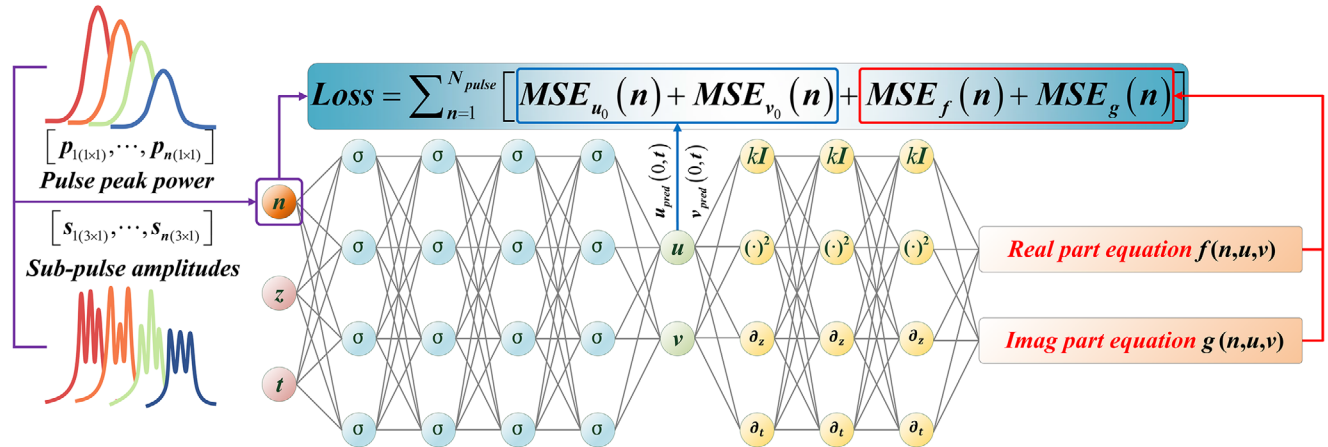


Figure 11. Generalizability modeling of PINN for scenarios with multipower/random waveforms in optical fibers.

simultaneously, the optimal model performance is achieved. In fact, when the physical constraints are stronger, the model performance will be further improved, and these constraints can be the exact data or prior knowledge such as symmetry conditions.

Next, we compared the model performance of PINN with the data-driven neural network (DDNN). The DDNN consists of four hidden layers with 100 neurons, and has the same network structure as PINN, where tanh and L-BFGS were adopted for activation and optimization, respectively. Similarly, DDNN took (z, t)

as input and (u, v) as output. In the constructed region, the exact data at equal distances were selected for data-driven modeling and as physical constraints of PINN. After the optimization of L-BFGS, DDNN fully fitted the exact data, while PINN also learned the governing equations with auxiliary coordinates at the same time. Without loss of generality, we recorded five sets of results with two methods, and calculated the average value as the final result. **Figure 15** shows the performance comparison between DDNN and PINN, where 20 000 auxiliary coordinates were

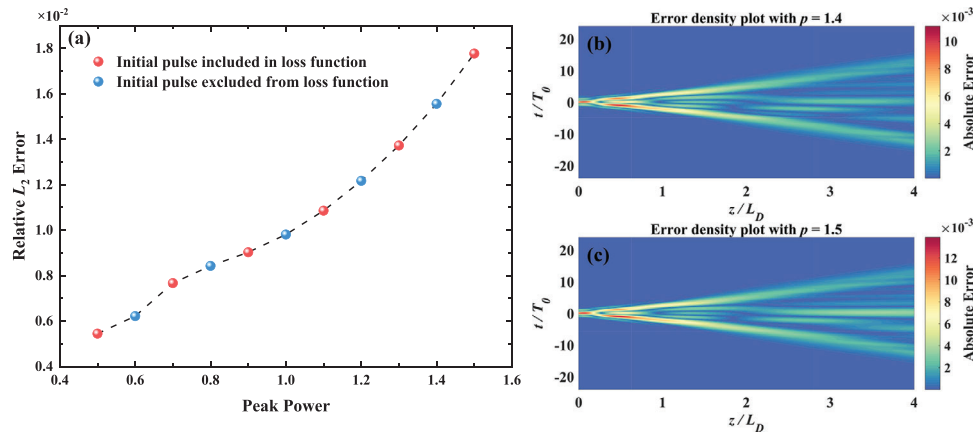


Figure 12. Generalizability demonstration: a) relative L_2 error of the cases with their corresponding initial pulses included in/excluded from the loss function, and error density plots with b) $p = 1.4$ and c) $p = 1.5$, respectively.

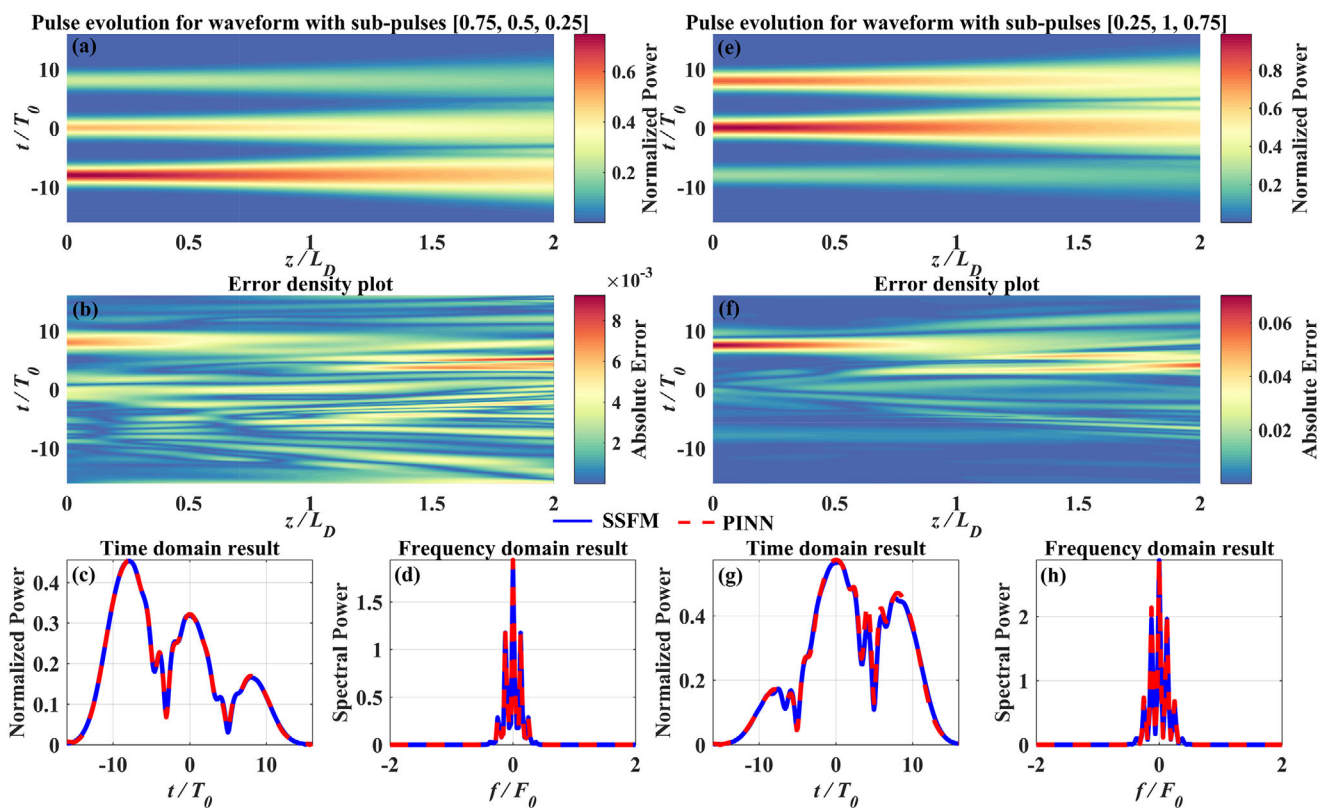


Figure 13. Modeling generalizability for random waveforms in optical fiber: a) pulse evolution generated by PINN, b) error density plot of PINN compared with SSFM, and c) time- and d) frequency-domain results for normalized and spectral powers at $2L_D$ for random waveform composed of 3 subpulses with amplitudes [0.75, 0.5, 0.25]. e) Pulse evolution generated by PINN, f) error density plot of PINN compared with SSFM and g) time- and h) frequency-domain results for normalized and spectral powers at $2L_D$ for random waveform composed of 3 subpulses with amplitudes [0.25, 1, 0.75].

used for PINN modeling. When the data volume is quite small, the error of DDNN is relatively large, and impossible to model pulse evolution. Specifically, DDNN requires the exact data of at least 5 pulses to characterize pulse evolution well. Furthermore, the performance of PINN is improved with the increase of data, which illustrates that more exact data contribute to better perfor-

mance of PINN. It should be noted that prior knowledge, such as the governing equations, is still the decisive factor affecting the model performance. This can be seen from the comparison between PINN and DDNN, i.e., even with a large volume of data, the performance of DDNN was still worse compared to PINN with only initial and received pulses as constraints.

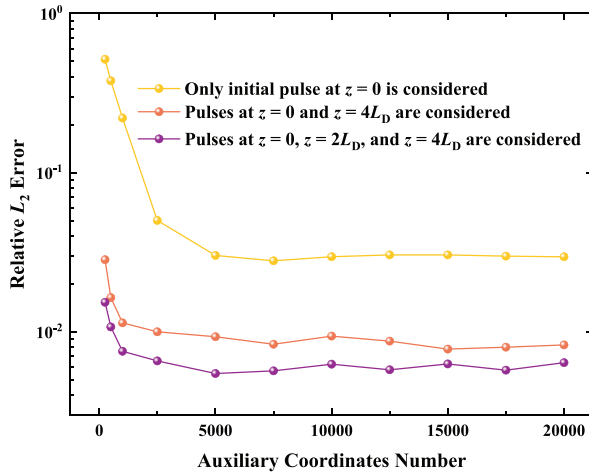


Figure 14. Relative L_2 error of PINN-based model with auxiliary coordinates number under several different physical constraints when GVD and SPM coexist.

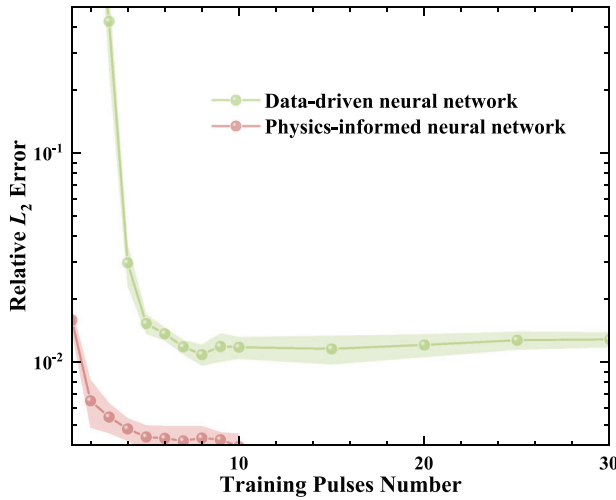


Figure 15. Relative L_2 error of PINN versus DDNN as a function of training pulses number.

4.2. PINN-Modeling Complexity Analysis

Finally, the comparative analysis on computational complexity of PINN and SSFM was conducted, where a theoretical calculation of multiplication operations in the modeling process is required. Such calculations are commonly applied to hardware complexity comparisons among different algorithms.^[37] For SSFM, the calculation performed in each step includes two fast Fourier transform (FFT) operations; each FFT operation contains $2N_{\text{FFT}}\log_2 N_{\text{FFT}}$ real-valued multiplications,^[38,39] where $N_{\text{FFT}} = N_0$ is the FFT size. Considering the entire pulse evolution process in the fiber and ignoring the exponential operation, the multiplications of SSFM can be expressed as

$$C_{\text{SSFM}} = N_{\text{step}} (4N_{\text{FFT}}\log_2 N_{\text{FFT}} + k_1 N_{\text{FFT}} + k_2 N_{\text{FFT}}^2) \quad (12)$$

where N_{step} denotes the number of steps into which the entire fiber is divided. This value must be properly set according to

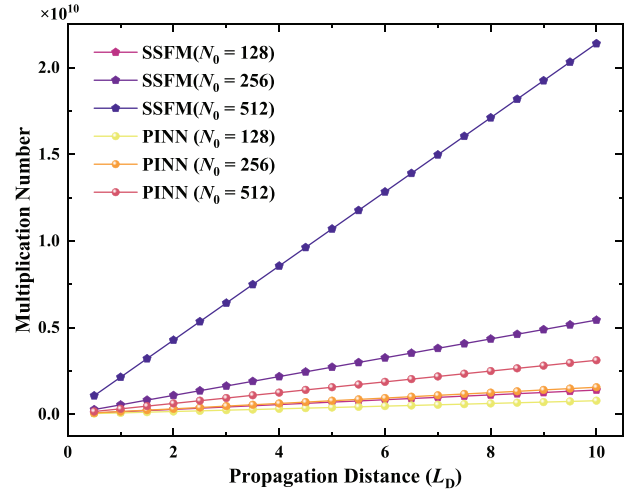


Figure 16. Comparison of computational complexity between SSFM and PINN.

the propagation distance and accuracy requirements. Further, k_1 and k_2 denote the number of rest multiplications proportional to N_{FFT} and N_{FFT}^2 in each step, respectively. From Equation (12), the SSFM complexity increases with the FFT size, and also, increases linearly with N_{step} .

The multiplication operation is also the main source of the PINN computational complexity. If the i th layer of the PINN network contains n_i neurons ($n_1 n_2 + n_2 n_3 + \dots + n_{(M-1)} n_M$) multiplications are required for a fully connected neural network with M layers. In addition, all terms in the loss function must be calculated for each coordinate. If the number of multiplication operations in this part is k_3 , the multiplications of PINN can be expressed as

$$C_{\text{PINN}} = (N_0 + N_f) (n_1 n_2 + n_2 n_3 + \dots + n_{M-1} n_M + k_3) \quad (13)$$

It is obvious from Equation (13) that the PINN complexity increases linearly with the number of selected coordinates, which is quite similar to the linear increase in SSFM with increasing steps. To facilitate a comparison of computational complexity of the two methods, calculations were performed for the same data volumes (i.e., $N_0 + N_f = N_{\text{FFT}} \times N_{\text{step}}$). For the two models, the relationship between the number of multiplications and propagation distance was plotted considering all parameters, as shown in **Figure 16**. An appropriate step size was set to ensure the accuracy of the two methods. Since the data volume was proportional to propagation distance, the number of multiplications increased linearly with propagation distance. Moreover, N_0 had far greater influence on SSFM complexity than that of PINN, as the former involved a large number of multiplication operations with $O(n^2)$ complexity in nonlinear calculations. When $N_0 = 512$, the multiplications of SSFM and PINN were $\approx 2.14 \times 10^{10}$ and 3.11×10^9 , respectively, for a distance of $10L_D$ (≈ 1000 km in a standard single-mode fiber). Thus, the computational complexity of the PINN-based model was only $\approx 14.6\%$ that of the SSFM-based model. In fact, the data volume that was actually required by PINN is not as much as SSFM, and the computational complexity of PINN can be even lower. Furthermore, the number of PINN multiplications is mainly affected by the neural

network structure rather than the data volume. Hence, PINN is more efficient in the case of large data volume.

5. Conclusion

In this study, we proposed a PINN-based solution to learn nonlinear dynamics in fiber optics, and carried out a systematic investigation and comprehensive verification on multiple physical effects including dispersion, SPM and higher-order nonlinear effects in optical fibers. The results demonstrated the strong characterization ability of PINN when simulating pulse evolution under multiple physical effects, and special cases including optical soliton and multipulse propagation. The peak power of initial pulse and amplitudes of subpulses in the random waveform were embedded in PINN as physical parameter controllers, which made the PINN learn the physical constraints corresponding to each scenario, thereby presenting robust generalizability in multiscenario modeling. Furthermore, the impact of auxiliary coordinates number and physical constraints on the PINN model performance was discussed and analyzed. In comparison with DDNN that uses the exact data dozens of times, PINN that only considers the initial and received pulse constraints acquired higher accuracy. Moreover, PINN generally requires far fewer multiplications than SSFM for the same data volume, especially for a far longer initial pulse. Therefore, the PINN studied in this paper fundamentally overcomes the limitations of SSFM and data-driven modeling methods, achieving a middle ground and incorporate the underlying physics in the NN learning for better modeling of nonlinear dynamics in fiber optics. This work aims at improving the generalizability of PINN and providing a key step for subsequent research on generalization modeling of PINN in future complex and practical scenarios. We believe that PINN has the great potential as a promising technique in our optics community for optical fiber communications, photonics material design, physical parameter identification, and optical signal processing.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

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automatic differentiation, fiber optics, nonlinear dynamics, nonlinear Schrödinger equation (NLSE), physics-informed neural network (PINN)

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